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AI and engineering careers: recent graduates' outlook on opportunities and challenges

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Abstract

Background The rapid advancement of artificial intelligence (AI) is reshaping industrial workflows and workforce expectations. After its breakthrough year in 2023, AI has become ubiquitous, yet no standardized approach exists for integrating AI into engineering and computer science undergraduate curricula. Recent graduates find themselves navigating evolving industry demands surrounding AI, often without formal preparation. The ways in which AI impacts their career decisions represent a critical perspective to support future students as graduates enter AI-friendly industries. Our work uses social cognitive career theory (SCCT) to qualitatively investigate how 14 recent engineering graduates working in a variety of industry sectors perceived the impact of AI on their careers and industries.

Results Given the rapid and ongoing evolution of AI, findings suggested that SCCT may have limited applicability until AI technology has matured further. Many recent graduates lacked prior exposure to or a clear understanding of AI and its relevance to their professional roles. The timing of direct, practical exposure to AI emerged as a key influence on how participants perceived AI's impact on their career decisions. Participants emphasized a need for more customizable undergraduate curricula to align with industry trends and individual interests related to AI. While many acknowledged AI's potential to enhance efficiency in data management and routine administrative tasks, they largely did not perceive AI as a direct threat to their core engineering functions. Instead, AI was viewed as a supplemental tool requiring critical oversight. Despite interest in AI's potential, most participants lacked the time or resources to independently pursue integrating AI into their professional roles. Broader concerns included ethical considerations, industry regulations, and the rapid pace of AI development.

Conclusions This exploratory work highlights an urgent need for collaboration between higher education and industry leaders to more effectively integrate direct, hands-on experience with AI into engineering education. A personalized, context-driven approach to teaching AI that emphasizes ethical considerations and domain-specific applications would help better prepare students for evolving workforce expectations by highlighting AI's relevance and limitations. This alignment would support more meaningful engagement with AI and empower future engineers to apply it responsibly and effectively in their fields.

Keywords Artificial intelligence, Engineering education, Undergraduate, Career, Social cognitive career theory, SCCT

Introduction

The rapid advancement of artificial intelligence (AI) has transformed technological landscapes, reshaping workforce expectations across industries. Engineering education has struggled to keep pace, leaving a growing curricular gap between what students expect, what industry demands, and what institutions currently provide (Akgun & Hosseini, 2025; Legatt, 2025). Whereas AI

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was once used primarily in highly specialized research, automation, and machine learning applications, since the advent of large language models (LLMs) and generative AI (GenAI), industries across multiple sectors are actively integrating AI technologies into their workflows, influencing job roles and required competencies as AI tools become embedded in technical work. In this manuscript, we use “AI” as an umbrella term, encompassing both traditional AI and generative AI, unless otherwise specified. While researchers have forecast that AI will “reshape the demand for expertise” and thereby change the nature of certain jobs, which jobs will be soon augmented by “human-AI teams” or which new expertise will emerge from this new AI-integrated workforce landscape remains unclear (National Academies of Sciences, Engineering, and Medicine, 2024).

Educational institutions must adapt their programs to prepare students for the unique challenges and demands of the AI-driven professional landscape. Recent scholarship highlights that educational institutions play a vital role in bridging the gap between academic training and professional requirements in the AI era (Călinescu & Tanaşciuc, 2024), and the emergence of AI as a distinct field in engineering necessitates a reevaluation of educational approaches and professional standards (Cao, 2022; Xu & Ouyang, 2022). For example, GenAI is particularly susceptible to bias and opaque reasoning, typically operating through complex, nontransparent algorithms (Marquis et al., 2024), which underscores the importance of equipping students with the skills to critically assess these systems and address their practical and ethical implications in real-world situations. Relatedly, a recent study of 90 graduate students found that adding an explicitly reflective module into a mandatory research ethics course significantly improved students’ awareness of ethical issues surrounding the development and use of AI (Usher & Barak, 2024); while the results from this graduate study are promising, extant literature does not address how to implement similarly effective ethics interventions in undergraduate engineering education at scale. Engineering experts echo this call for a proactive response to integrating AI ethically and effectively into professional engineering preparation nationwide (American Society for Engineering Education [ASEE], 2024).

While some universities incorporate AI-related coursework, we found no unified approach in literature for teaching engineering students how to integrate, critically evaluate, or challenge the conclusions of AI-driven tools in technical work. Integrating AI into engineering curricula is crucial, offering opportunities for personalized learning and enhanced simulation engagement while preparing students for AI-integrated workplaces (Mosly, 2024; Xu & Ouyang, 2022). Universities are exploring

ways to incorporate AI-related coursework, as research has underscored the necessity of adapting curricula to prepare students for evolving job landscapes (Tucker et al., 2020), but there is no standardized approach yet (Tuovinen & Rohunen, 2021; Walter, 2024). A systematic review of research published 2011–2021 found that the effectiveness of AI integration into science, technology, engineering, and mathematics (STEM) education at scale relies heavily on the available technological resources at institutions and the pedagogical preparation of instructors in applying AI technology (Xu & Ouyang, 2022). Exploring the effective integration of AI and its impact on STEM and engineering education remains an important emerging trend in the literature (Chiu et al., 2025; Xu & Ouyang, 2022). Despite these efforts, current curricula at engineering universities regarding AI are lagging behind industry’s development and expectations surrounding AI use, and even among universities, the extent to which students are exposed to theoretical and hands-on knowledge of AI in industry-relevant applications is highly varied.

Recent engineering graduates represent a key demographic of interest that is navigating these evolving industry expectations and keenly experiencing these gaps in education. The school-to-work transition presents challenges beyond technical competencies, involving organizational socialization and adapting to new cultural contexts (Lutz & Paretti, 2021). Unlike established professionals who may be adapting to AI later in their careers, recent graduates are entering the workforce at a time when AI adoption is accelerating. The variation in preparation of students across different institutions may lead to varying levels of familiarity, confidence, and preparedness for AI-related tasks and navigating ethical concerns in industry roles. Research on how engineers work across different disciplines and professional roles shows that real-world engineering involves complex sociotechnical skills and that there is a disconnect between what students learn in their engineering programs and what they actually need to thrive in the engineering workplace (Jesiek et al., 2017).

Despite this, scant research has addressed how recent engineering graduates perceive AI’s impact on their careers and industries. Existing research primarily has focused on AI’s technical capabilities; the ethics surrounding its use; and its impact on the educational system, the economy, and the environment, but to the best of our knowledge, few studies have explored how new engineers view and experience these advancements firsthand. In one exception, a study surveying 715 new engineers found that workers in engineering specialties that involved critical decision-making showed less fear that AI would eliminate their jobs (Khogali & Mekid, 2024),

but a knowledge gap regarding the expectations, concerns, and preparedness of early-career engineers in response to AI's growing role in engineering professions remains. Conducting interviews with recent engineering graduates offers firsthand perspectives and insights into how these technological advancements have influenced their career-making decisions.

In this qualitative study, we utilize semi-structured interview responses from 14 recent undergraduate engineering participants across various industry sectors to address two research questions:

RQ1: How do recent engineering graduates perceive academic and career interests, learning experiences, outcome expectations, and self-efficacy beliefs related to AI?

RQ2: How do recent engineering graduates perceive the implications of the future impact of AI on their industries?

Background

AI and higher education

What is AI education? Our concept of it here encompasses both traditional AI and GenAI. Traditional AI refers to systems that use rules, logic, and statistical models to perform a myriad of tasks through analyzing existing data (e.g., optimization, recommender systems, predictive modeling); its roots trace to the 1950s. GenAI refers to newer systems—such as LLMs or image generators—that are capable of producing new content (e.g., images, text, code) based on a set of training data. ChatGPT and other LLMs became popular on the global stage in 2023 and have become the most visible examples, but they represent just one category of GenAI tools, which also include image, audio, and code generators. While GenAI is a subdomain of AI, its practical and ethical implications are distinct, especially as it pertains to its unique concerns around “hallucinating” or confidently fabricating false results (Metz & Weise, 2025). While many AI developers have engineering backgrounds, our work focuses on the more general concept of AI competency for engineers. It aligns with U.S. Exec. Order No. 14277 (2025), which calls for 1) training engineers who can create the next generation of AI models and 2) providing engineers AI competency, or “the foundational knowledge and skills necessary to adapt to and thrive in an increasingly digital society.” Engineers with AI education, we suggest, can adopt state-of-the-art AI technologies to improve their engineering careers. It may include learning how to use types of GenAI like LLMs to summarize documentation, write code for data analysis, or to streamline general administrative tasks, but it might not

teach students how to design and implement new language models or the foundational theories of machine learning.

In a hopeful sign for AI education, the integration of AI into STEM education has been a positive development generally. A growing body of literature documents the multifaceted benefits of AI-enhanced learning environments, including in relation to critical factors for fostering career readiness such as fostering student engagement and interest in STEM disciplines, boosting self-efficacy, and improving learning outcomes (Xu & Ouyang, 2022). Studies of K–12 environments show that implementing hands-on AI-integrated learning activities enhance students' interests and self-efficacy in STEM careers, including those related to AI (Aliabadi, 2023; Tsao & Tsao, 2023). Likewise, recent scholarship in higher education has shown that AI-integrated learning environments can enhance students' engagement and self-efficacy in STEM (S. Wang et al., 2023; Wu et al., 2023) and creativity (S. Wang et al., 2023). Students' use of AI tools, particularly GenAI tools like ChatGPT, improve their confidence about employability and job market readiness (Xiao & Zheng, 2025). AI-enhanced online personalized adaptive learning tools have been designed for use at historically Black colleges and universities (Dong et al., 2024); future research will describe the success of implementation. Similarly, digital twin-based teaching and learning in engineering programs has been used as a vehicle to prepare students for Industry 4.0 and 5.0, highlighting the potential for AI-augmented simulation environments to foster hands-on, practice-based learning (Hazrat et al., 2023). Dependence on AI among STEM undergraduates influences their research self-efficacy and productivity (Buniel et al., 2025), which indicates that regular AI use may strengthen students' confidence in academic and professional contexts. As the speed at which AI is being embedded into engineering workflows accelerates, the need to adapt our methods in our classrooms to reflect the affordances of these tools has grown.

The rise of AI in industry is already pushing universities to reconsider a traditional emphasis on theoretical instruction, which has often come at the expense of experiential and practical learning that students value (Adorni & Grosso, 2025). Engineering students themselves increasingly want more comprehensive AI-related ethics training in their curricula (Goldenkoff & Cech, 2024), and some institutions have implemented AI literacy modules, focusing on training students to ethically use and critically evaluate GenAI tools like ChatGPT (Feldman & Cherry, 2024; Walter, 2024). The availability of opportunities for students to engage with AI through hands-on experiences remains unevenly distributed

(Božić, 2023). A recent study of public sentiment found a largely positive or neutral attitude towards AI in STEM education, suggesting that the public is more likely to support curricular changes that encourage its use in STEM education (Smith-Mutegi et al., 2025). The California State University system has launched the largest AI deployment in higher education, integrating AI tools and training across its 23 campuses (Fitzpatrick, 2025; Watanabe, 2025). Research calls for similar large-scale intentional curricular shifts toward applied AI training, prioritizing project-based learning and collaborative labs over lectures and preparing students for modern professional engineering environments requires such shifts (Mosly, 2024). However, Rigler (2016) noted that engineering curricula tend to be slow to adapt to emerging technologies, often because of lengthy institutional approval processes—a problem that worsens as technology changes more quickly.

At the same time GenAI is invading the classroom in two ways: first, through student use. Students are using GenAI models for assistance with reading and writing tasks, including proofreading, summarization, and drafting assignments (Cooper, 2023) as well as to produce functional code and solve problems in many introductory computing courses (Denny et al., 2024; Kazemitabaar et al., 2023). Second, through instructor use. Canvas, a popular learning management system, recently announced that GenAI features will be available to “help with grading, scheduling, generating rubrics and summarizing discussion posts” (Palmer, 2025). However, while these uses may constitute students learning to use AI to be effective engineering students or instructors using it to be effective instructors, they do not constitute students learning to use it for engineering tasks in the workplace.

Preparing students to use AI in the workplace will involve preparing instructors to give them such preparation. Instructors frequently lack a background in using AI and feel underprepared to incorporate AI tools and software into their teaching. Many resist the concept (Dewan et al., 2025), and educators tend to hesitate to adopt new technologies with which they are not familiar in the classroom generally (Rigler, 2016). Some instructors prefer traditional teaching methods, avoid altering established pedagogical approaches, and can see AI integration as a threat to their jobs (Bohara & Rana, 2024; Fitria, 2023). As the use cases of AI and GenAI tools continue to increase, many scholars have begun to argue for a balanced approach to layering in GenAI-driven enhancements in engineering education, citing the facts that AI remains an emerging technology in certain engineering fields and that overreliance on GenAI tools may weaken students’ deep understanding of traditional engineering

principles (Cañavate et al., 2025). This balanced approach that cautiously integrates aspects of AI into engineering education acknowledges that failing to educate students on the practical applications of AI tools will put them at a disadvantage in the modern job market (Broo et al., 2022).

AI and industry

AI is rapidly becoming a core competency in the engineering workforce. Industry leaders consistently identify AI proficiency as a key hiring criterion (Blochinger et al., 2024). The increasing integration of AI and GenAI in engineering industries and embedding of AI tools in technical work has transformed traditional engineering practices, leading to a growing demand for AI-skilled engineers who are proficient in AI-related competencies, including automation, machine learning, digital twin technology, and AI-enhanced design processes (Zia et al., 2024). This includes tasks such as code generation and text refinement (Brachman et al., 2024), and as a result AI applications now permeate marketing, project management, software development, engineering design, and other core business functions. Even in those roles where AI adoption is less extreme (e.g. core engineering functions such as robotic process automation and decision support), the integration of AI technologies is significantly impacting job roles and required competencies by eliminating human involvement in some routine tasks (Stephany, 2024). At the same time, we know little about engineers’ potential concern about job security because of AI integration, a problem that was profound in a study of AI integration in an international call center in one study (Presbitero & Teng-Calleja, 2022), despite engineers’ far more specialized expertise compared to call center workers. A recent study of 4,867 developers at major tech companies found that less experienced software developers had higher adoption rates of a company-supplied AI-based coding assistant and greater productivity gains (Cui et al., 2025), showing that integrating AI-based tools can be a good investment for companies. At the same time, there is growing concern about the collapse of early-career pipelines for this same reason, as the cost of AI coding tools is significantly lower than junior developer salaries, which may incentivize companies to replace novice coders with AI agents (Dave, 2025).

AI is also creating new job opportunities. This is particularly true in fields like machine learning and data science (Tschang & Almirall, 2021). For example, with Google announcing that more than 25% of its new code is AI generated, experts have suggested that the focus of computer science education should shift toward software engineering practices that use a collaborative model where students focus more on quality assurance and

integration and less on coding syntax and generation (Talagala, 2024). Meanwhile, in fields like product design, digital twin technology is revolutionizing the way engineers develop and test prototypes, using AI to simulate real-world conditions (Hazrat et al., 2023), and engineering firms are integrating AI into structural simulations and real-time monitoring systems (Waqar, 2024). For example, a recent comprehensive bibliometric analysis of AI-based decision support systems in construction demonstrated these systems' technical potential in optimizing project scheduling, risk management, and resource allocation (Waqar, 2024), though it stops short of assessing how these AI-enhanced systems might influence human decision-making. It is thus crucial that academic training align with industry needs, lest graduates struggle to apply AI modes in real-world engineering environments (Ahmad, 2020).

As industry needs continue to rapidly evolve, gaps may only grow between engineering education and employer expectations, and these concerns are not new. Employers have historically expressed dissatisfaction with graduates' preparedness for the workplace, emphasizing deficiencies in communication and time management, and failure to engage in continuous learning (Kajale & Shaikh, 2021; Ramadi et al., 2016). Employers report that new engineering hires often lack practical experience in key AI applications, including AI model deployment, industrial automation, and real-time AI decision-making (Blochinger et al., 2024). Though industries anticipate that future engineers will be able to work with AI-powered design tools, universities have been slow to incorporate them into their curricula (Patel, 2025; Waqar, 2024).

There remains substantial uncertainty in higher education about how to prepare engineering students for future interactions with AI in a professional setting. Experts in engineering education have recognized that educators must respond to the proliferation of AI in the classroom and in the workplace, but they have offered few suggestions for achieving this at scale (ASEE, 2024). One potential solution is to support industry-sponsored AI projects, AI-enhanced coursework, and AI certification programs that can provide students with the skills industry is demanding (Dong et al., 2024; Savelka et al., 2025), but the impact of these AI-enhanced projects on students' career trajectories and decision-making processes is not yet known.

Existing literature does not adequately address AI's career-related implications for early-career engineers, despite how research has highlighted the relevance of AI in industry careers. We lack rich insights into how engineers perceive their readiness to engage with AI and GenAI tools, and how that readiness affects their career-related decisions. Our work addresses existing gaps in

the literature by investigating the perspectives of recently graduated engineers working in industry and extending AI-in-education research further into the workforce domain by illuminating how early-career engineers interpret the role of AI in shaping their professional identities and career pathways. This exploratory study seeks to provide additional guidance about what would benefit students by exploring how newly graduated engineers have interacted with AI technology in the workforce and how it has influenced their career plans moving forward, so we can better understand the modern challenges engineering students face related to AI.

Guiding theoretical frameworks

Social cognitive career theory

Because we are primarily interested in career decision-making, we grounded our work in social cognitive career theory (SCCT). Based on Bandura's social cognitive theory, SCCT addresses the career decision-making process through the lens of intricately linked factors that influence a person's career choices and how those factors interact (Lent et al., 1994). SCCT has been aptly described as a "comprehensive career theory that argues that an individual's career path results from the interaction between multiple career elements" (Wang et al., 2022, p. 2). Research has shown that SCCT is applicable to academic and career development processes across life stages (Lent et al., 2002). Over the last several decades, numerous researchers have utilized SCCT with a diverse range of students and professionals at different stages of their development (Fouad et al., 2010; Lent et al., 2002), and it has been specifically used to study students, including ones from underrepresented groups, in STEM disciplines such as engineering and computing (e.g., Fouad et al., 2010; Lee et al., 2015; Lent et al., 2005, 2008, 2013; Navarro et al., 2007).

We focused our analysis on the components of SCCT that were most salient in our data: learning experiences, outcome expectations, interests, self-efficacy beliefs, and contextual influences proximal to choice behavior. Access to learning opportunities—as determined by a variety of factors, including but not limited to educational institution and personal characteristics—subsequently affects learning experiences. These experiences in turn influence self-efficacy and outcome expectations, which can change a student's career interests and goals. *Self-efficacy* is a dynamic set of beliefs about oneself that individuals hold toward certain domains or tasks. It is defined by the belief that one is well equipped to succeed within a given domain or at a certain task (Lent et al., 1994). One can have high self-efficacy in a particular domain but lower self-efficacy for another domain. For example, an engineering student may have high self-efficacy for coding

but lower self-efficacy for working on a team. *Outcome expectations* refer to one's imagined consequences of a specific behavior—that is, what one believes will happen if they follow a particular course of action (Lent et al., 1994). An example of this would be “If I took Job X, I would get higher pay, but I would be less interested in the work.” Outcome expectations generally reflect one's learning experiences; for example, if you have had positive outcomes doing a similar task, you may expect a similar positive outcome again.

Learning experiences, outcome expectations, and self-efficacy jointly determine students' interest in a given career. SCCT defines career interests as “patterns of likes, dislikes, and indifferences regarding career-relevant activities and occupations” (Lent et al., 1994). These theoretical constructs are tightly interwoven in an ongoing feedback loop, as people's interests in activities tend to grow when they believe they can perform well and anticipate positive results. As that interest grows, people then choose and set new goals to maintain or expand their engagement in the activity. Their continued participation then provides new successes or setbacks to inform their learning experiences and subsequent interests in continuing to engage in the activity (Greenhaus & Callanan, 2006). Contextual influences can then support or hinder a student's decision about a particular career. These are influences that are proximal to choice behavior that directly or indirectly affect one's behavior at critical choice points (Lent et al., 1994). Examples would include having a large professional network that grants better access to certain careers or discriminatory hiring practices that prevent a student from obtaining a particular job.

Despite the theory's extensive use in STEM disciplines, most studies using SCCT to study new technologies, such as virtual reality and AI, have focused on how the technologies can be used as learning tools rather than their influence on career choice (Burhan, 2025; Jiang et al., 2021; Thisgaard & Makransky, 2017). A U.S.-based study using a qualitative case study design found that many undergraduates viewed using AI tools as an inevitable element of their future workplace and expressed an intention to learn about these tools as a supplement to their existing skillsets (Colón-Aguirre & Bright, 2025). In addition, a handful of studies show that exposure to AI tools can shape students' career interests, aspirations, and self-efficacy in technology-related pursuits. A cross-sectional study by Duong and colleagues (2024) that integrated SCCT with the theory of planned behavior found that undergraduates students who frequently used ChatGPT showed an uptick in interest in technology-related entrepreneurial careers. Similarly, a large-scale quantitative study in

Indonesia (N=698) found that including AI tools (e.g., AI-enhanced accounting software, spreadsheets, and visualization tools) in undergraduate accounting curricula enhances students' AI literacy and perceptions of workforce readiness in terms of AI self-efficacy (Maulana et al., 2025).

To the best of our knowledge, this study is the first to use SCCT as a framework to understand the impacts of AI and exposure to AI education in undergraduate curriculum on early career engineers' decision-making processes. Through rich descriptions of participants' perspectives, we seek to provide a broad understanding of the phenomenon under study.

Affordance theory as a consideration

During analysis, we introduced Norman's adaptation of Gibson's affordance theory to address the impact of the historical timing of our interviews (Gibson, 1979; Norman, 1988). We conceptualized this study in early 2020; Chat GPT became publicly accessible in late 2022, which led to the declaration of 2023 as GenAI's breakthrough year (Singla et al., 2025); our interviews occurred in 2023–2024. Norman's adaptation of affordance theory focuses on how people perceive possibilities for action in a particular environment or technology and how they consequently act on those perceived possibilities. Applying this adaptation to the fields of design and human–computer interaction, Norman advocated designing objects and digital interfaces in a way that clearly communicates their intended purposes and capabilities so that users can envision possible affordances more easily.

In the context of this study, AI's affordances would refer to early career engineers' perceptions of what AI would enable or constrain for them in their current professional position. The key aspect of affordance theory relevant to this work is that a variety of factors, not just inherent capabilities, shape a tool's perceived affordances, including the user's goals, prior knowledge, cultural beliefs, past experiences and context of use (Norman, 1988, 1999). For instance, a person lacking prior experience with the advanced features of Microsoft Excel may perceive its primary affordances as performing calculations; others may realize it can be used to automate tasks through macros or perceive the usages of dynamic data dashboards. Generally, the more aware a person is of the potential use cases of a technology in their field or position, the more likely they will be able to articulate or envision what that technology may enable them to do. Given the emergent nature of AI and GenAI at the time of the interviews, we comment on how this timing may have shaped participants' responses through the lens of affordance theory in our methods section.

Methods

We used a qualitative research design grounded in a constructivist epistemology. Constructivism recognizes how participants make meaning of the world around them (Guba & Lincoln, 1994; Patton, 2015). We interviewed a total of 14 participants using a semi-structured interview guide aligned with SCCT.

Quality in qualitative research

We applied Walther et al.'s (2013) quality in interpretive engineering education research, or Q3, quality framework in all aspects of designing and implementing our study. The first phase of the Q3 approach, “making the data,” includes the process of planning for, sampling, and collecting qualitative data. Making the data is directed to the “internal customer” for example, the participants and research team. “Handling the data” includes every aspect from transcription to analysis and interpretation with careful documentation of decisions, steps, and procedures. Handling the data is directed toward the “external customer” for example, the research community, broader public, and research team (Walther et al., 2013, p. 638). The framework consists of process reliability and five types of validation: theoretical, procedural, communicative, pragmatic, and ethical (ethical validation was added to the original framework in 2018 by Sochacka et al.). Throughout our description of our methods, we parenthetically indicate how our research design and choices align with the Q3 framework's measures of quality. Additionally, our research methods were reviewed and approved by our universities' Institutional Review Boards.

Authors positionality

We drew on Secules and colleagues' (2021) recommendations for considering contextualized positionality using their orienting reflection questions (p. 36). We will share how our interests, motivations, and prior experience with AI impacted how we chose to approach this research and how they may impact what we were able to observe as researchers. At the time that Julie and Paul initially conceptualized this project in early 2020, they were motivated by the increasing need for engineering curricula to incorporate AI-related courses and the apparent lack of research to explain how students felt AI would affect their career choices in a shifting job landscape. Little did we know then how ubiquitous generative AI tools would become over the course of the project, nor did we anticipate how the project team would evolve over time in its composition of expertise.

We thoughtfully considered our “insider/outsider status” per Secules and colleagues' recommendation (2021,

p. 36) with respect to two identities most salient to this work: engineering education research (EER) and familiarity with AI. Through our reflections, we realized these are not binary statuses but a continuum, and that many of us have shifted in the continuum over the course of the project. Throughout the project, Julie's and Paul's statuses have been opposite and complementary, with Julie having over 20 years of EER expertise and little desire to seek out or use AI tools beyond what is integrated into digital platforms she regularly uses (e.g., AI-enhanced search engines). Meanwhile, Paul runs an AI-driven robotic biomedical engineering laboratory as part of his research agenda and had no prior EER experience before starting this project. As such, Julie guided the team's major methodological decisions while Paul complemented this with his insider knowledge of AI to inform the development of the interview protocol. Our postdoctoral and graduate student researchers who later joined the team represented varying combinations along these spectra as well, with some having more or less experience conducting EER or integrating AI tools into their workflows. One of the intended outcomes of this work was for all team members to grow their familiarity in both EER and AI over the course of the work through near peer mentoring, as documented in Martin and colleagues (2024).

Interestingly, the wide variety in insider/outsider status in EER and AI familiarity among the members of the research team served as an unintended quality measure, as those less familiar with AI were able to critically challenge the data analysis by asking big-picture questions on the exact intent of participants' statements during peer debriefs. Similarly, those who were more EER “insiders” constantly checked the others' interpretations against their prior EER experiences and knowledge of SCCT literature. Our varied familiarity with AI allowed us to approach the project with an openness about how it was being used by working engineers in the field. We all had engaged in ethics discussions surrounding AI use in higher education and were curious to hear participants' industry perspectives. As the doctoral trainees on the project, Isaac, Landon, and Deepthi move toward being insiders on both spectra, it has prompted critical discussions that resulted in developing ethical guidelines for the appropriate use and limitations of AI in research and academic contexts.

Making the data

Participant recruitment/selection

We employed a purposeful sampling strategy by recruiting participants via LinkedIn and chain sampling (Patton, 2014). Prospective participants completed a brief survey that indicated their undergraduate degree and current professional role. The resulting data gave us the

context for their responses and ensured that interview participants met our inclusion criteria of having a job as an engineer in industry for 1–5 years. We interviewed 14 participants who met the criteria and compensated them with a \$50 check. This sample size is appropriate for an exploratory study (Braun & Clarke, 2021); other researchers (e.g., Guest et al., 2006; Hennink & Kaiser, 2022) have found that saturation generally occurs with 12–13 interviews, and that themes can emerge within six interviews. We completed five interviews in June 2023 (Participants 1–5) and nine in June 2024 (Participants 6–14). As shown in Table 1, the sample offered a broad range of experience in engineering programs and industry, with eight engineering subfields (plus a second major in STEM education) and 11 industries represented.

Semi-structured interviews

One member of the research team, Deepthi, conducted 45- to 60-min semi-structured interviews via Zoom videoconferencing platform (process reliability). Questions addressed the participants' (1) background and academic preparation; (2) general career goals and expectations; and (3) AI-related experiences, career goals, and expectations. While the interview questions aligned with our research questions and SCCT (theoretical, procedural, and communication validation), we avoided the use of theoretical and research jargon in interviews so that participants would construct their own meaning about the influence of learning experiences, environmental influences, and other aspects of their academic and career decisions (communicative and pragmatic validation).

We developed the interview protocol through discussion with our project's expert advisory board members and piloted the interview guide during the first few interviews, making minor changes based on participants' initial responses (communicative validity). The interview guide contained three sections that queried participants' (1) educational background before entering the workforce, (2) field of work and company description (3) views of AI in their field. We used the critical incident technique (Grant & Trenor, 2010) to direct participants to think about specific examples illustrating their responses (communicative and procedural validation). Interview questions aligned with various aspects of SCCT in order to answer RQ1 regarding recent engineering graduates' perceptions of academic and career interests, learning experiences, outcome expectations, and self-efficacy beliefs related to AI and RQ2 regarding their perceptions of the implications of the future impact of AI on their industries (procedural and theoretical validation) (examples of interview questions are provided in Table 2). We recorded the interviews and had them professionally transcribed (procedural validation).

Handling the data

We read and corrected transcripts while listening to the recorded interviews (procedural validation and process reliability) and maintained confidentiality by scrubbing the transcripts of identifiable information (ethical validation). We then used MAXQDATM software to conduct inductive and deductive thematic coding (Braun & Clarke, 2006) with a priori codes related to the constructs

Table 1 Participant demographic information

ID	B.S. grad year*	Interview	Undergraduate major	Current sector of work
1	2019	June 2023	Materials Science & Engineering	Research and development
2	2015	June 2023	Civil Engineering	Water infrastructure
3	2021	June 2023	Electrical Engineering	Aerospace, space, and defense
4	2019	June 2023	Bioengineering	Healthcare
5	2020	June 2023	Computer Science	Nonprofit educational services
6	2021	June 2024	Bioengineering	Pharmaceutical production
7	2023	June 2024	Chemical Engineering	Industrial gases
8	2022	June 2024	Bioengineering	Medical devices
9	2020	June 2024	Civil Engineering	Climate technology
10	2020	June 2024	Mechanical Engineering	Environmental engineering
11	2018	June 2024	Biomedical Engineering and STEM Education	Manufacturing, design, work-force development
12	2020	June 2024	Plastics and Composites Engineering	Plastics
13	2020	June 2024	Mechanical Engineering	Manufacturing
14	2020	June 2024	Industrial and Systems Engineering	Banking

*Note that this is the year in which the interviewees reported earning their bachelor of science degrees; this is not necessarily the same as the year they entered their current industry, as some pursued graduate degrees and others changed fields. Each interviewee had between 1 and 5 years in industry, collectively

Table 2 Sample questions from interview guide and their link to the theoretical framework and research questions

Section	Sample questions	Link to SCCT	Contributes to RQ
Educational background	Can you tell me about your undergraduate education? Tell me about your most interesting courses What was the least interesting thing you studied? Tell me about any technical electives you took.	Learning experiences	RQ1
Field of work and company	What does your company do? What type of work do you do in a typical day?	Contextual influences	RQ1
Views of AI in their field	How important do you think AI is in your field right now? How do you think AI will impact your field in the future? When/how did you become interested in AI? What did you learn about AI in your undergraduate studies? What did you learn about AI after graduating? How is your career different because of AI? Have you had any exposure to AI, either as an undergraduate or at work? Do you consider yourself to work in an AI field? If not, do you have any interest in switching to an AI career? How would you learn about AI now? Would your career be different if you worked with AI? How?	Learning experiences, outcome expectations, self-efficacy, contextual influences, choice goals and actions	RQ1, RQ2

of SCCT along with emergent codes from the exploratory data (Patton, 2015; process reliability; pragmatic, theoretical, and communicative validation). For example, we coded for aspects of SCCT outlined in the guiding theoretical frameworks section and identified variation in the ways that different participants described their experiences that related to the a priori code “outcome expectations,” related to AI careers (see Table 3 for more information). Two research team members, Isaac and Landon, first trained together before independently coded each transcript and negotiated differences to consensus through a series of discussions (procedural and theoretical validation, process reliability). As the coding and negotiation phases progressed, the rest of the team periodically discussed emergent findings and interpretations with them and made suggestions for expanding or collapsing individual codes (communicative validation, procedural and theoretical validation, process reliability). After coding was complete, the entire research team met multiple times to discuss and challenge our collective understanding (process reliability; communicative, pragmatic, and theoretical validation).

Considerations related to interview timing

Due to the timing of the interviews, our early career engineer participants may not have been able to strongly articulate perceived affordances of AI or GenAI. One reason for that is that media outlets had begun to use the terms AI and GenAI interchangeably (Sand Technologies, 2025). As well, the integration of GenAI directly into daily workflows was still relatively new at the time. On the other hand, some AI-enabled features in engineering software may have gone unrecognized due to their

ubiquity. Early career engineers such as our study participants may have never encountered engineering software without embedded traditional AI elements. For example, the number of firms that had adopted traditional AI in 2023 was 55%; however, by 2024, 72% of firms had incorporated AI into operations, an increase spurred in part by a 22% increase in the number of firms using GenAI tools (Singla et al., 2025). Further complicating our analysis of interviews at that particular time, specific interpretations of GenAI’s usefulness may not have existed widely at the time of the interviews, further limiting perceived potential affordances of GenAI (i.e., participants may have found it difficult to envision specific applications of GenAI to their daily tasks). In contrast, GenAI tools such as LLMs have tended to be more visible and user-facing, inviting more direct interaction in user’s natural languages. These GenAI tools have dominated much of the media’s conversation around AI in the past few years and have been shown to be more frequently applied to STEM learning contexts than other AI technologies that have less public visibility such as computer vision (Chng et al., 2023).

Any difficulty in articulating AI’s affordances may be further compounded by the confusion caused by falsely conflating GenAI with AI. Some participants may have had little exposure to AI in general and may have incorrectly assumed that the questions in the study related only to GenAI. They may have opinions about traditional AI’s affordances they did not think to share with the interviewer. Participants having fully informed opinions on AI and GenAI tools’ usefulness in assisting specific tasks in their professional role directly impacts the affordances they can perceive from AI and GenAI. Our study’s

Table 3 A priori codes (SCCT dimensions), their definitions, and themes and outcomes that arose from coding

SCCT dimension	Definition	Themes and outcomes
Learning experiences	Prior personal, academic, or vicarious experiences that shape self-efficacy and outcome expectations, influencing future interest and career goals	Participants' exposure to AI varied widely, with many encountering it for the first time in college or post-graduation. These experiences influenced their baseline familiarity but were often limited to theoretical rather than hands-on applications
Self-efficacy beliefs	One's confidence in their ability to successfully perform specific tasks or roles	Many participants expressed low confidence in their ability to work with AI, citing limited technical background and lack of exposure, which led to hesitation or avoidance of AI-related career paths
Outcome expectations	Beliefs about the likely results of performing particular actions, such as anticipated rewards, challenges, or consequences tied to a given career choice	Participants held a range of expectations, from AI being essential for handling complex data to being a helpful tool but not transformative. These beliefs shaped how they envisioned AI's role in their current and future work
Career interests	Patterns of attraction or aversion toward specific activities or occupational fields, shaped by learning experiences, self-efficacy, and outcome expectations	Interest in AI tended to grow through meaningful, hands-on experiences, though some participants saw AI as peripheral to their main career goals due to perceived complexity or time constraints
Choice actions	Observable behaviors or steps taken to pursue particular educational or career paths, such as enrolling in courses, applying for jobs, or using certain tools	Participants who were interested or felt some self-efficacy reported taking concrete steps like using AI tools at work or exploring online learning. Others showed interest but were stalled by time, confidence, or access issues
Choice goals	Intended future outcomes or aspirations, such as achieving a particular role or skill set, that guide decision-making and motivate sustained engagement	Some participants aspired to integrate AI into their careers or pursue additional credentials in the future, while others were content in their current roles and saw AI as an optional enhancement rather than a necessary direction
Contextual influences (proximal)	Environmental factors (supports or barriers) that occur at key decision points and can directly influence whether and how an individual acts on their interests or goals	Participants noted that future engineers would likely be influenced by broader societal shifts such as AI-related ethical concerns, job displacement, regulatory uncertainty, and hype cycles, even if these didn't shape their own decisions

findings reflect this complexity and the need for future AI education to explicitly address both traditional AI and GenAI going forward.

Complementary frameworks might have allowed us to explore participants' behaviors and tendencies in making technology adoption decisions in greater detail, such as the technology adoption model (Davis, 1989) and Rogers' (2003) diffusion of innovation theory, but exploring these were beyond the scope of the current study. For instance, Roger's (2003) diffusion of innovation theory characterizes users (e.g., innovators, early adopters, early majority, late majority, laggards) based on when they choose to adopt certain technologies compared to the technology's percentage of market share (Rogers, 2003); however, it implies a relatively linear adoption process that may not fully capture the complexities of how professionals encounter emerging technologies in fast-changing or ill-defined domains. In many cases, participants expressed uncertainty about the value, relevance, or reliability of AI and GenAI tools, suggesting a lack of perceived affordances, rather than a lack of interest or innovation orientation. Thus, using affordance theory as an interpretive lens allows us to better center the relational, contextual nature of AI engagement and open up richer interpretations of how engineers learn about and navigate emerging tools in practice.

Results and discussion

In this section, we will first discuss our findings on how recent engineering graduates perceived AI in relation to personal impact and the broader impact on society. We define *personal impact* as how participants perceived or anticipated that AI would influence their professional careers. Here grounding our findings within SCCT offers insights into the decision-making processes of engineering participants in how they chose or did not choose to integrate AI into their evolving career path. We define *broader impact on society* as how participants anticipated shifts in contextual influences that would influence career choices related to AI for engineers in the future. This theme had several emergent subthemes (ethical considerations, efficiency, regulatory and privacy concerns, rapid growth) that offer insight into factors that future participants may consider as they decide whether to pursue an AI-related career.

Personal impact

Certain *learning experiences*, specifically, exposure to AI in professional or self-guided contexts, and the timing of these experiences seemed to be especially influential in participants' decisions regarding their careers. For example, Participant 7 encountered AI for the first

time through their undergraduate studies. They felt that curricular options related to AI should be tailored and marketed such that students can explore areas of personal or career interest in a more targeted manner. Initial exposures in structured educational settings helped Participant 7 develop proficiency in understanding programming languages. They observed:

[I]f [a student] wasn't really interested in doing [AI] as a chemical engineer, they would probably hate [machine learning courses], so I think just the fact that maybe if you make it [clear] that these other AI courses are available and [students] can self-select to enroll in them [it would be better], because I don't think there's anything [in AI] that was really necessary in my job.

By facilitating targeted career discovery, Participant 7 suggested, AI education would allow participants to explore various applications of AI, helping them identify which AI-related courses and projects they find most engaging. This approach would help participants clarify their career interests, allowing them to make informed decisions about integrating AI into their future roles and enhancing the alignment of their evolving skills with their career aspirations.

Outcome expectations related to AI adoption revealed diverse participant perspectives on how AI would impact their professional growth, daily tasks, and broader career trajectories. Participants voiced varied expectations, often shaped by personal experience with AI in workplace settings. For example, Participant 4 shared that observing AI applications in the manufacturing company where they work made them realize how vital it was to remain competitive. They said that for a manufacturing plant to "survive,"

you need AI to look at all of your different process parameters, because you've got sensors every two feet for your entire process across eight stories of manufacturing. So there's tens of thousands of real-time inputs coming in, and no human can manage all of that at once. So there are a lot of the process controls that are done using AI.

This illustrates how direct exposure to AI fosters a more nuanced understanding of its capabilities and relevance to real-world applications. Participant 4's outcome expectations that AI can manage vast data inputs beyond human capability shaped their actions, as individuals come to see AI as essential for thriving in complex, data-intensive environments like manufacturing.

Direct exposure to AI's capabilities thus fosters a nuanced understanding of its importance, aligning expectations with practical applications across fields.

Participant 2 reflected a cautious stance toward AI, saying they would

use it if I have to. If it's a tool that's presented, I'll use it. I'm not a person that's afraid of the technology. I tend to [use] caution because I don't like to be too dependent on things that may prove unreliable at some point.

Participant 3 adopted a more pragmatic stance when they commented,

I have interest in working with AI. I don't know if I have interest fully in being an AI engineer. That work is hard. And right now I'm happy with the work, the style of work I'm doing, especially doing embedded stuff. It's very cool. It blends my hardware and software backgrounds [electrical engineering] very well. I think having the information about the AI makes my job a lot easier, if I'm going to be able to work with it, because I know exactly what I'm doing.

While still recognizing AI's value, these individuals held outcome expectations that positioned AI as supportive rather than foundational to their professional career trajectory or acknowledged potential shortcomings of AI use.

In summary, the participants' varied outcome expectations demonstrate a spectrum of beliefs about AI's role in their professional futures, encompassing aspirations for enhanced skill sets, caution over current capabilities, and selective integration based on career relevance. These findings reflect the broader implications of SCCT, where outcome expectations contribute to shaping engagement, readiness, and perspectives toward adopting emerging technologies like AI.

Another strong influence that contributes to choice actions and goals related to integrating AI in career paths is *self-efficacy beliefs*. As expected, whether participants believed they could engage effectively with AI influenced their choice to do so. For example, Participant 10 noted:

When I was in college, I barely knew about AI. I didn't even know what ChatGPT was. I found out about that a year and a half, two years ago. So I was pretty behind on any of that stuff. I don't think we really ever used AI a whole lot [in my classes].

They said, further, that they believed there were other students “who knew what was going on,” who had a stronger background in AI and its applications.

Participant 13 had low self-efficacy not because of a lack of exposure but because of a perceived lack of aptitude. They noted, “I personally don't have any desire to get into [AI] only because that involves programming, and I am not very good with programming.” Their

self-doubt underscores how difficult technologies can challenge self-efficacy, leading participants to question their ability to engage with advanced tools even if they might otherwise be interested in the field.

Other participants saw AI as complex and were unsure if they would have the opportunity to learn it. Participant 1 said, of their interest in moving toward a career in AI,

At my current level of understanding and education, I couldn't because I just don't understand it. But if the opportunity presented itself to me where I could easily obtain a master's degree in AI and biology or something and if I could pair that with my current career trajectory, I totally would, because I think that that would add a lot to my skill set and I think that that's where the field is going.

Participant 7 observed that AI was not yet “really common in my industry, but I think just the fact that people are interested in learning more about it and seeing the value is a step that is pretty interesting to me.” This response reflects an awareness of a potential skills gap, where the evolving needs of the industry create uncertainties in participants about their current AI capabilities. However, there is an understanding that AI can expand their skill set, and the growing interest is a positive development.

Interests, as affective responses to activities or domains, are influenced by factors such as self-efficacy beliefs, outcome expectations, and individual goals. In our study, participants' interest in AI often developed through real-world applications and hands-on experiences. Participant 4 noted that working on AI projects illuminated the value of AI in their field, bioengineering, stating,

I don't think in three days you can discover a new cancer therapy, but just even proving the effectiveness of AI for at least narrowing down your list of candidates to test at in vitro for cancer research was really cool. So that was definitely my most hands-on AI work I've ever done. It was really helpful for just getting an understanding of what [AI] is and what it isn't.

Consistent with SCCT literature, this quote suggests that direct, practical exposure to a field can deepen interests and inform career paths. Such applied experiences appear pivotal in solidifying participants' interest in AI, illustrating the importance of hands-on learning opportunities.

While AI sparked interest for Participant 4, Participant 10 saw it as more peripheral to her core career aspirations. In response to a question about whether they are interested in switching to an AI-based career, Participant 10 stated,

No. I really like where I'm at. I think AI can be a great help and I look forward to seeing what changes happen with it and what gets developed. But no, I like where I am, and I like working more in a civil engineering aspect.

Others saw AI as more of a tool to improve production and value at their companies. Participant 14, who works in the finance world, stated that AI was “important” and had a lot of “value ... to improve our technology stack [meaning at the company where he works] and improve the product that we have.” Participant 11 highlighted challenges balancing AI learning with other responsibilities, remarking,

I've seen so many of those [online courses] for AI and all of that and I didn't enroll in them, but I wanted to. I was always like, “Oh, I would love to enroll in one of those Coursera courses.” There's just so many online certifications now, I was like, “Wow, there's a lot of stuff,” so just haven't had the time. I looked into it.

Participant 11 seemed to feel interest in learning AI but did not feel sufficient pressure to do so to put other obligations aside.

Recent engineering graduates perceived their interests in AI often emerged through real-world, hands-on applications where they could utilize AI to streamline their own jobs. Outcome expectations varied, from seeing AI as essential complex environments to treating it as a helpful tool. Self-efficacy beliefs, especially among those with limited prior exposure or who had low self-efficacy beliefs, significantly influenced the participants' willingness to engage with AI, ultimately affecting how they envisioned integrating AI into their careers.

Broader impact on society

While our SCCT framing of this study focused primarily on how individuals chose to make career decisions, we began to notice several emerging factors (ethical considerations, efficiency, regulatory and privacy concerns, and rapid growth of the field) that participants felt would influence the career decisions of their successors—i.e., new graduates in a world where AI has increasing presence and impact in daily workflows. These emergent subthemes represent *proximal environmental influences* that engineering graduates may consider as they decide whether to pursue an AI-related career.

Overall, the participants' discussions revealed significant concerns about the large-scale societal effects of AI. One of their concerns centered around job security. For instance, Participant 10 reflected that “the big worry for everyone” was whether AI would replace them. Similarly,

Participant 2 showed concern about our society's willingness to allow AI to replace unskilled labor positions, saying,

I'm not trying to make excuses for human inefficiencies, but there's a level of that element, that social element that could really be detrimental to people who don't have PhDs or master's degrees. There's a lot of unskilled labor that's going to go away. I don't know, I kind of like the aspect of unskilled labor. I don't like the image of machines picking things or machines doing every single thing; interacting with people, even though if it's unskilled labor or middle-tier labor, that's what makes us a society cohesive.

This sentiment captures apprehensions regarding AI's implications for employment.

Further, Participant 10, whose current work is in the field of environmental engineering, noted the limitations of AI in handling nuanced, human-centered aspects of engineering, such as adapting plans to individual needs. They were skeptical. “There's a human component to this job that I don't really think AI can replicate,” he said. Participant 9 said, “I use AI in some capacity, as long as it's not to the point where it's reframing my ideas because we do have to produce our own work.” This response indicates a belief that it is unethical to allow AI to take on certain professional responsibilities of an engineer. This belief influenced participants' broader attitudes toward the integration of AI in their fields. Through SCCT's lens, the participants' reflections highlight how societal concerns and human values influence career-related decisions, motivating participants to consider the ethical and practical aspects of AI adoption. These values may drive career paths where human discretion remains essential, reflecting an orientation toward responsible, ethically aware AI integration.

Participants emphasized the transformative *efficiency* of AI in streamlining repetitive or administrative tasks, allowing them to focus on more intellectually demanding responsibilities. For example, Participant 8 suggested that AI could allow engineers to dedicate more time to projects that demand higher-order problem-solving skills and reduce “all the other little tedious tasks,” thereby increasing “efficiency.” Similarly, Participant 6 stated, “I think it's really important to help reduce effort on the learning part. If you need something done really quickly, but you don't have time to take a full course that may have a much more thorough teaching regimen or coursework.... I use ChatGPT to help me as a shortcut.” They added that they had used ChatGPT to help them learn about Microsoft Office, Excel, and coding in HTML and JavaScript. This suggests the integration of AI can reduce mental strain

and project fatigue by addressing tasks that traditionally hinder time allocation toward creativity and critical thinking. Participant 11 mentioned using ChatGPT to expedite the production of written content and assist with design tasks, underscoring AI's capacity to simplify multi-tasking in real time: "I use ChatGPT ... to check all of the things I'm writing, help me out a little bit, creating different types of content that I need to publish. It also helps me with the design of different things," which suggests that AI not only offers practical support but also influences quality control by providing feedback that enhances clarity and coherence. This points to AI's potential role as a valuable collaborator, one that can reinforce output quality as much as it boosts productivity.

Participant 4 said that they thought that project management tools that are currently "pretty stupid" will soon be AI-assisted, that integrations will mean that "common tasks" that "typically take [a certain] amount of time" will be "schedul[ed] and handl[ed]" better. In other words, they felt AI has significant potential to optimize time management, enhance task tracking, and integrate predictive tools that anticipate project demands, illustrating a belief that AI can evolve from simple task automation into proactive, resource-optimizing functions. According to SCCT, the perception of growing efficiency in professional roles with the help of AI can directly impact career decisions, engineering graduates may perceive that AI-related career paths will leverage AI in ways that reduce the menial workload, allowing them more time to engage in higher-order thinking and creative activities necessary for engineering innovation (Baltà-Salvador et al., 2025). In the long term, engineering graduates may find themselves drawn toward roles that align more closely with a more technologically enhanced, streamlined workflow, thereby influencing their career aspirations and choices.

Even with its improved efficiencies and streamlined workflows, participants voiced significant *regulatory and privacy concerns* regarding the limited transparency of AI. They were concerned that many AI models operate as "black boxes" such that humans have limited insight into how the model works or how proprietary or otherwise sensitive data is handled. This aligns with existing research emphasizing the need for AI models to foster trustworthiness through making their development process transparent to reveal potential biases (Barnes & Hutson, 2024). Participant 3 voiced their concerns thus:

The problem is you can't have tech companies self-regulate themselves, because they're not going to do that. As a company, [self-regulation is] I guess

what you want, because it makes you the most amount of money. But I do think there needs to be some government oversight.... I think it's more of an act now, and you can always tweak it along the way as opposed to wait and see what happens. But there's two perspectives. One is from a corporate standpoint, and one is from a government standpoint.

Participants were also concerned about how proprietary or sensitive data is exposed to AI systems. For instance, Participant 3 said:

I know that privacy is huge, and even across sectors, like Samsung had a huge code base leak, because the software engineers were just pushing code to public AI models, and then a lot of the codebase got leaked. And I feel like I hear that now at every company.

This highlights apprehensions about AI's vulnerability to data breaches, a major concern for many participants, who anticipated such lapses could affect their work environments. Making participants aware of these ethical implications of AI use could guide future participants toward careers that advocate for responsible AI use or specialties that focus on engineering policy or regulatory compliance.

Lastly, many participants mentioned the *rapid growth* of the AI field and the surrounding "hype," which make some participants feel obligated to learn how to use AI because of its widespread popularity. Participant 12 stated, "I don't have much of a knowledge base of AI.... I know it's getting bigger. So I wanted to get a little bit of an understanding of it." Some participants also had little fear about how AI would change society. Participant 13, for example, said:

I'm not scared of [AI], only because even if it does start taking jobs, they are jobs that nobody really wanted to do in the first place. That's what I feel like is going to be the case.... Technology advances, and technology always has something to replace. Before, we used to have people on assembly line making the Ford cars. Now, they have robots doing the same thing.

This reflects concerns about mismatches between AI's anticipated and actual capabilities. This perception aligns with studies showing that the intense media portrayal of AI capabilities can fuel both excitement and skepticism within the general public (Natale & Ballatore, 2017). Participant 13 also had a fatalistic view about AI: "I think there's a ton of potential for it. I see it's going to only grow. There's no stopping it in my opinion." This statement aligns with research indicating that the fast-paced

development in AI often overwhelms practitioners of AI (Huang & Rust, 2021). This overwhelming pace may affect participants' career development by fostering resilience but also contributing to self-efficacy challenges as they navigate a fast-evolving field. This duality could shape participants' engagement with AI or create barriers if users' expectations of AI and its perceived capabilities diverge too widely.

Given their thoughts about broader impacts, we see that recent engineering graduates anticipate that AI will significantly reshape their industries, with implications that future professionals will need to consider this when making career decisions. With concerns such as workforce displacement, ethical risks, regulatory challenges, and AI's current inability to replicate nuanced human judgment, participants emphasized the need for responsible innovation and oversight, but they recognized AI's growing role in society, boosting productivity and transforming workflows. With the fast-paced development and surrounding hype sparking a mix of enthusiasm and pressure to adapt, the perspective of engineering graduates will be an important one in shaping AI's role in our society.

Implications and limitations

Implications of the findings

Our work has implications for both the practice and policy regarding the integration of AI into engineering education. To ensure engineering graduates are well-equipped for the evolving role of AI in the workforce, educational institutions and industry leaders must work together to enhance AI integration in curricula, foster stronger industry partnerships, and establish clear workplace guidelines.

Our study indicates that engineering curricula must include AI education and explicitly address the practical and ethical implications of traditional AI tools and GenAI tools. Study participants widely agreed that early exposure to AI tools and its implications would better prepare graduates for the evolving discourse surrounding the appropriateness and applicability of different AI tools in the workforce. A novel finding of this study is that early career practitioners are not readily equipped to adapt to AI integration in the workplace, in much the same way as mid- or late-career engineers face a substantial adjustment; even without having a lengthy baseline experience working in industry, early career practitioners overwhelmingly recognize the inevitability of integrating AI and GenAI tools in their industries and anticipate a variety of difficulties and wide-ranging impacts to their fields and society at large. They called for AI education that would prioritize practical, hands-on learning experiences, allowing students to engage with real-world

projects, internships, or case studies that are AI-related. With this approach students will develop the confidence and expertise needed to thrive in professional settings. By working directly with both AI and GenAI tools, students can gain more familiarity with each tool type's affordances and limitations, making them better prepared to meet industry demands.

Even when prompted, participants had few thoughts about how the presence of AI influenced their career choice process (i.e., influence on choice actions, choice goals, and performance domains related to AI use). This may reflect a lack of sufficient knowledge of or experience with these tools, limiting the potential affordances they could perceive as being directly tied and beneficial to their career goals. Essentially, traditional AI-enhanced features of common engineering or professional software might have been too ubiquitous for our participants to recognize as AI, and GenAI may have been too emergent for them to have fully informed opinions about its capabilities or usefulness as it related to specific tasks related to their professional roles as engineers.

Future work conducted later in the lifespan of GenAI might not suffer from this problem; until then, integrating affordance theory into the initial design of an SCCT-guided study might capture the influence of rapidly emerging technologies on students' career decisions. It may have been premature to apply SCCT to GenAI at the time of the interviews, and as the technology approaches its market share saturation level, studies using SCCT may more robustly capture its impact on students' career choice processes. History suggests this was the case with exploratory SCCT-grounded studies about technological developments in the past, such as data science and cybersecurity (Rogers, 2003). However, framing perceived affordances directly as proximal contextual influences within SCCT in the early design of a study could prove to be an effective way to adapt SCCT to more directly describe the impacts of a rapidly emerging technology on students' decision-making processes.

To maximize the impact of AI education, our findings suggest a personalized, context-driven approach that allows students to see AI's applications in their respective domains will make learning more meaningful and applicable to their careers. At the same time, all undergraduate engineering students should study ethical AI use, bias mitigation, data privacy, and transparency. The resulting knowledge will help students navigate practical and ethical challenges in their future careers, enhance their confidence in using AI, and reduce the gap between academic learning and workplace expectations regardless of the direction they take (Abulibdeh et al., 2024). This study also underscores the importance of tailoring AI curricula to allow all graduates to develop a well-rounded

perspective on AI. This could be in the form of courses and workshops that focus on AI's societal impact. Addressing the broader implications of AI in society can also help students become responsible engineers who consider ethical, legal, and economic consequences when designing AI-driven solutions.

Given AI's varied applications across different industries, sector-specific guidelines should be developed to address the unique ethical and operational considerations within each field (Vatankhah et al., 2024). Even outside of industry, policymakers in higher education have yet to agree on a unified AI policy with respect to scientific research and publication practices (Nam & Bai, 2023), and we recommend that both industry and higher education focus on developing responsible use policies as they relate to AI. Additionally, AI "upskilling" initiatives should be encouraged within workplaces, allowing employees to continuously refine their AI competencies in response to evolving industry standards (Morandini et al., 2023). By setting clear expectations and providing ongoing training, employers can create a structured, responsible approach to AI adoption in the workplace. Accrediting bodies, e.g. ABET, also have a role in defining the role of AI in engineering. Accreditation standards need to be updated to reflect the ethical concerns when licensed engineers adopt AI to assist with their work. These concerns are specific to the profession of engineering and may be overshadowed by education's own ethical concerns with the adoption of AI for teaching and learning.

Notably, some participants held misconceptions about the skills required to engage with AI, with one (Participant 13) believing that a lack of programming ability precluded AI learning. This reveals a deeper educational gap, not only in clarifying the accessibility of modern AI tools but also in fostering a growth mindset around technical competencies. Institutions must therefore go beyond teaching content, actively working to challenge fixed mindsets and empower students to see themselves as capable learners in emerging technological domains.

Keeping pace with the rapid growth of AI requires flexible, continuous learning opportunities. Online tutorials, certifications, and self-paced courses should be made available to ensure students and professionals alike can stay updated on AI advancements. These flexible formats also make AI education more accessible to a broader range of participants, including those in nontraditional learning paths such as working professional seeking career transitions, self-taught learners, individuals without formal higher education, and those in vocational or technical training programs.

As in other aspects of engineering education, fostering a culture of lifelong learning and skill development

will be crucial in helping engineering graduates adapt to the fast-paced evolution of AI. Institutions can offer certifications or ongoing training programs that allow participants to continue enhancing their AI skills beyond graduation. This will help them be competitive in the workforce and prepared for its demands. Beyond certifications, direct engagement with industry professionals is essential. Guest lectures by AI experts, industry mentorship programs, and real-world case studies integrated into coursework provide students with firsthand insights into AI's challenges and opportunities in engineering. Internship and co-op opportunities focused on AI applications offer another avenue for students to gain practical experience before entering the workforce. Furthermore, joint research initiatives between universities and industry partners will strengthen this connection by enabling students to work on AI projects that contribute to real-world industrial advancements (Esangbedo et al., 2023). In this way universities can ensure that students are not only well-versed in AI theory but also equipped with the skills and professional exposure necessary for success in AI-integrated workplaces. We note, however, that industry professionals are themselves grappling with the impact as AI technology progresses and may not know how to best incorporate AI into their workplaces. We believe this uncertainty requires closer connections between industry professionals and educators as they navigate the shared experiences of the AI revolution.

Limitations

This study has several limitations that should be considered when interpreting the findings. First, while all 14 recent engineering graduates were actively working in industry, the majority were not employed in AI-related fields, which limited the range of performance domains and attainments they could discuss. At the same time, the findings give a broad sense of recent graduates' feelings about AI. While our sample delimitation was purposeful due to funding resources, future work could include current senior-level students who are currently navigating career choices amidst rapid AI and GenAI development. We collected limited demographic information about the participants; as a result, we cannot draw conclusions based on those trends.

As this was an exploratory study, findings may not be transferable to other contexts, and there is a need for a broader and more diverse sample. Lastly, during analysis, we noticed that several participants—particularly those interviewed before AI became widely popular in mid-to-late 2023—lacked a clear and sufficient understanding of what AI is, leading to responses not specifically oriented toward AI technologies.

Conclusions and future work

The integration of AI into engineering education is not just a curricular enhancement; it is an inevitable strategic imperative for preparing future engineers to meet the evolving demands of the modern, technology-driven workforce. This exploratory study underscored the importance of embedding hands-on, experiential AI learning into academic programs and the necessity for ethical grounding and context-specific applications of AI in students' field. To bridge the gap between theory and practice, universities should integrate AI into their engineering programs in a way that combines technical proficiency, interdisciplinary relevance, and ethical awareness. This will ensure that students will graduate not only capable of interacting effectively with AI in their professional roles, but also as responsible users of the technology.

Equally important is the role of industry partnerships to inform workplace readiness efforts in this evolving landscape, especially since industries can often adapt collaborative partnerships between academia and industry can provide students with critical exposure to real-world applications of AI in their field through internships, certification programs (e.g., microcredentials), and mentorships. Our findings suggest that recent graduates are interested in the potential of AI to improve efficiency in certain roles, yet they currently lack the resources or bandwidth to learn to integrate AI into their professional roles. Therefore, employers should support engineering graduates' transitions to the workforce by establishing clear criteria and guidelines for ethical AI usage and offering ongoing training or "upskilling" opportunities related to AI. Future work might include a quantitative phase that would allow us to draw conclusions about various work sectors and nature of different engineering tasks with respect to AI usage, as this was beyond the scope of the current study. By working together, higher education and industry leaders can cultivate an engineering workforce that is highly adaptable, ethically grounded, and fully equipped to harness the transformative power of AI in engineering practice.

Our theoretical contribution is to extend the use of SCCT to capture students' career choice process as it relates to breakthrough new technology (AI), as the technology continues to be rapidly integrated into industry workflows. Because most of our interviewees became new users of AI tools between 2023 and 2025, some of them lacked sufficient experience with these tools to articulate meaningful career decisions and goals centered around AI use. As a result, our findings speak primarily to one portion of the SCCT framework (learning experiences, outcome expectations, interests, self-efficacy beliefs, and contextual influences proximal to choice behavior). In related future work, we believe that

integrating affordance theory with SCCT at the research design phase would allow for a more robust investigation of the influence of rapidly emerging technologies on students' decision-making processes. Additionally, we posit that the technology adoption model could be a strong complementary framework, which posits that individuals' acceptance and use of a given technology hinges on their intentions to do so, which are subsequently determined by the users' perception of the technology's usefulness and ease of use (Davis, 1989). With emergent technologies like AI, these alternative framework combinations may provide a clearer lens through which to interpret the potential impact of future proximal influences on students' career decisions related to AI compared to using SCCT by itself.

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The use of AI in manuscript preparation

The authors confirm that generative AI tools were not used in any way the conceptualization, writing, or editing of this manuscript. All aspects of the manuscript reflect the authors' own work.

Author contributions

JM and PJ contributed significantly to the project conceptualization, funding acquisition, methodology, and project supervision. JB contributed to data curation, data analysis, project administration, and supervised the graduate students on the project. ID and LS contributed significantly to data analysis and coding. DS conducted the research interviews. All authors contributed to writing pieces of the manuscript, and all authors read and approved the final manuscript.

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Availability of data and materials

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Human ethics and consent to participate

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Competing interests

The authors declare no competing interests.

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